

1 Shared ESS Planning Framework

1.1 Architecture

Algorithm 1: dummy

$\mathbf{x}$

1.2 Master

$$a \tag{1}$$

$$b \tag{2}$$

The two layers are coupled through the investment plan $\mathbf{x}$ only. The master layer is a derivative-free search over the lattice of admissible plans: it uses the value $Q(\mathbf{x})$ returned by the recourse and nothing else. No gradient, dual multiplier or cut is passed upward, because $Q(\mathbf{x})$ is the value of a certified consensus point of a nonconvex problem (Subsection 2.2.7) and is neither convex nor smooth in $\mathbf{x}$; the cut-based master of an earlier version of this work was abandoned for that reason. To represent long-term storage ageing without sacrificing decomposition across SOs, the operational layer includes a dedicated *shared ESS agent* that tracks State-of-Health (SoH) and the resulting evolution of available energy capacity. This preserves decomposition of the TSO and DSO subproblems across representative years while maintaining intertemporal consistency of ESS degradation over the planning horizon.

Algorithm 2 is a mesh adaptive direct search (MADS) on granular variables $[\cdot, \cdot, \cdot]$. Every candidate is a point of the investment lattice: converter power in multiples of Δ^S , energy in multiples of Δ^E , within the duration, capacity and budget constraints of Subsection 2.2, which are enforced in closed form before any evaluation. Each evaluation is one complete run of the operational recourse (Algorithm 1, A.1), from a cold start and under one frozen configuration, so that $Q(\mathbf{x})$ is a deterministic function of $\mathbf{x}$ and the cache is exact. The search terminates when the poll at unit poll size fails, that is, when a positive spanning set of lattice neighbours of the incumbent has been evaluated and none improves it by a determinate margin, or when the evaluation budget is exhausted, in which case the incumbent is reported with

Algorithm 2: Derivative-free shared-ESS planning search (mesh adaptive direct search on the investment lattice) with certified ADMM recourse evaluations

Input: Candidate nodes E^S ; representative years Y , days D and scenarios Ω_M, Ω_O with probabilities; lattice units Δ^S (MVA) and Δ^E (MWh); duration bounds $\phi^{\text{Min}}, \phi^{\text{Max}}$; capacity limit $E_{e,y}^{\text{Max}}$; budget c^{Inv} ; initial poll size Δ_0^p ; evaluation budget N^{max} ; certification parameters of Subsection 2.2.7; direction seed.

Output: Incumbent plan $\mathbf{x}^*$, its certified value $F(\mathbf{x}^*)$, the poll history and the evaluated neighbourhood with determinacy verdicts.

// Admissible set, enforced in closed form before any evaluation

$\mathcal{X} = \{\mathbf{x} = (S_{e,y}^{\text{Inv}}, E_{e,y}^{\text{Inv}})_{e \in E^S, y \in Y} : S_{e,y}^{\text{Inv}} \in \Delta^S \mathbb{Z}_{\geq 0}, E_{e,y}^{\text{Inv}} \in \Delta^E \mathbb{Z}_{\geq 0}, (1)-(2)\}$

Scale: $\mathbf{z} = \text{diag}(\Delta^S, \Delta^E, \dots)^{-1} \mathbf{x}$, so that $\mathcal{X}$ is a subset of $\mathbb{Z}^n$ and the finest mesh is the lattice itself

Cache $\mathcal{C} \leftarrow \emptyset$ (one certified evaluation per canonical plan; a cache hit is never re-evaluated)

// Initial search step: a designed set of lattice points (single-node ladders, combinations, staging)

Evaluate the designed set with Algorithm 1 and the certification rule; add the records to $\mathcal{C}$

$\mathbf{x}^{\text{inc}} \leftarrow \arg \min_{\mathcal{C}} F$; $\Delta^p \leftarrow \Delta_0^p$; $N \leftarrow |\mathcal{C}|$

while $N \leq N^{\text{max}}$ **do**

 // Poll step

 Generate the $2n$ OrthoMADS directions $\pm \mathbf{v}_1, \dots, \pm \mathbf{v}_n$ from the next Halton point of the recorded sequence

 Poll set $\mathcal{P} \leftarrow \{\mathbf{z}^{\text{inc}} + \Delta^p \mathbf{v}\}$, each point rounded to the lattice and, if it violates (1)–(2), snapped to the nearest admissible lattice point; drop duplicates and cached points

if $|\mathcal{P}| < n + 1$ **then**

 Add to $\mathcal{P}$ the admissible lattice points within one unit step of $\mathbf{z}^{\text{inc}}$ in the ∞ -norm (completion set), up to the stated cap

 Evaluate every point of $\mathcal{P}$ (full poll, in batches of the available concurrent slots); an evaluation that does not certify, or whose storage agent is infeasible (SoH floor), is recorded with its cause and assigned $F = +\infty$ (extreme barrier); add the records to $\mathcal{C}$;

$N \leftarrow N + |\mathcal{P}|$

if some $\mathbf{x} \in \mathcal{P}$ has $F(\mathbf{x}) < F(\mathbf{x}^{\text{inc}})$ by a determinate margin (Subsection 2.2.7) **then**

$\mathbf{x}^{\text{inc}} \leftarrow$ the best such point; $\Delta^p \leftarrow 2\Delta^p$

else

if $\Delta^p = 1$ **then terminate:** $\mathbf{x}^{\text{inc}}$ is mesh-locally optimal on the lattice

the poll size reached. The method therefore claims mesh-local optimality on the lattice at the stated resolution, and no global optimality gap: the convergence theory of MADS applies to F as a function on the lattice with the extreme barrier and does not require continuity of Q , which is why a direct search was preferred to gradient- or cut-based masters for this recourse.

The master problem minimizes the expected discounted investment cost plus the certified expected operating cost of the plan:

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{X}} F(\mathbf{x}) &= I(\mathbf{x}) + Q(\mathbf{x}), \\ I(\mathbf{x}) &= \sum_{e \in E^S} \sum_{y \in Y} \sum_{c \in \Omega_C} \frac{\omega_c}{(1+r)^{y-y_0}} (c_{y,c}^{\text{Inv},S} S_{e,y}^{\text{Inv}} + c_{y,c}^{\text{Inv},E} E_{e,y}^{\text{Inv}}) \end{aligned} \quad (3)$$

where $c_{y,c}^{\text{Inv},S}$ and $c_{y,c}^{\text{Inv},E}$ are the unit investment costs of converter power rating and energy capacity under cost trajectory c , $S_{e,y}^{\text{Inv}}$ and $E_{e,y}^{\text{Inv}}$ are the investment decisions, and $Q(\mathbf{x})$ is the expected discounted operating cost of the plan returned by the recourse (Subsection 2.2.7). $I(\mathbf{x})$ is closed-form and cheap; every evaluation cost is in $Q(\mathbf{x})$. The terminal (salvage) value of the installed assets is not part of $F(\mathbf{x})$: operation does not internalize it, and it is reported beside every figure as a post-evaluated credit computed from the terminal SoH and the remaining calendar life (Section 4).

Investments are sized in commercial units: $S_{e,y}^{\text{Inv}}$ is a non-negative integer multiple of Δ^S and $E_{e,y}^{\text{Inv}}$ of Δ^E , and $S_{e,y}^{\text{Inv}} = 0$ implies $E_{e,y}^{\text{Inv}} = 0$. The values used in the case study are $\Delta^S = 0.25$ MVA and $\Delta^E = 0.5$ MWh with $\phi^{\text{Min}} = 2$ h and $\phi^{\text{Max}} = 4$ h (Section 3).

For each candidate plan, the scenario-independent investment decisions $S_{e,y}^{\text{Inv}}$ and $E_{e,y}^{\text{Inv}}$ are passed to the operational recourse as fixed parameters; no probability-weighted averaging of mutually exclusive physical ESS capacities is performed. The recourse returns the certified value $Q(\mathbf{x})$ and its certification record; nothing else flows upward.

1.2.1 Recourse Evaluation and Certification

The recourse value of a plan is the expected discounted operating cost of the coupled system over the horizon,

$$Q(\mathbf{x}) = \sum_{y \in Y} \frac{Y_y}{(1+r)^{y-y_0}} \sum_{d \in D} D_d \sum_{m \in \Omega_M} \sum_{o \in \Omega_O} \omega_m \omega_o C_{y,d,m,o}^{\text{Op}}(\mathbf{x}) \quad (4)$$

where $C_{y,d,m,o}^{\text{Op}}$ is the daily operating cost of representative day d of representative year y under market scenario m and operation scenario o at the consensus point reached by the coordination procedure of A: generation cost in the TN, activated flexibility and curtailment in the DNs, and, in the multi-scenario instance, the commitment-deviation charge of Subsection 2.3. Settlement flows between operators are transfers and are excluded from $Q(\mathbf{x})$; they are reported separately. Each representative year stands for a block of Y_y calendar years and is discounted at its representative year.

Stopping and certification. Each recourse evaluation runs the consensus ADMM of A until the primal and dual residuals of every consensus channel pass the absolute-plus- relative test of Boyd et al. [?], with $\varepsilon_{\text{abs}} = 10^{-5}$ and $\varepsilon_{\text{rel}} = 10^{-4}$, in a cycle in which every local NLP solve is successful. Passing the residual test does not imply that the objective has settled: on the reference evaluations of the case study the objective moved by a further 15–21 k€ after their residual-based certification. The residual pass at cycle k_0 therefore only opens the certifying regime, which is then held fixed: acceleration off, the tight interior-point tail on, and the penalty parameters ρ frozen (A). An evaluation is certified at the first cycle $k^* > k_0$ at which all of the following hold:

1. the objective has shown at least three turning points since k_0 , so that its period $\hat{P}$ (measured between the first and third turning points) is measured rather than assumed;
2. the successive half-swings of the objective are not growing; swings smaller than $\tau/10$ are treated as noise and are neither compared nor counted as turning points;
3. the range of the objective over the last $W = \max\{20, \lceil 1.1 \hat{P} \rceil\}$ cycles is at most τ ;
4. the priced interface-consensus gap — the value of the residual mismatch between the operators’ interface schedules at the interface price — is at most $\tau/2$ in absolute value;

5. every cycle in that window was solved cleanly. A local solve is clean when it ends at the solver’s optimal status, or at its acceptable status on a primary attempt with all four termination metrics within ten times the tight-tail tolerances.

An objective that descends without oscillating is certified by a monotone branch when, over a window of $2P_{\max}$ cycles with $P_{\max}$ the longest period measured on the instance, its steps are decreasing, its range is at most τ , and $|\text{last step}| \times 2P_{\max} \leq \tau$, a linear bound on the remaining descent that uses no fit. An evaluation that satisfies neither branch within its cycle cap is reported as uncertified, with the objective’s movement since the residual pass (its settling slack) and its consensus gap.

Resolution. The threshold τ is set from the value resolution the paper claims: $\tau = \delta_R V/4$ with $\delta_R = 0.07$ and V the storage value of the reference unit in force when the rule was adopted ($V = 259,375.33$ €, hence $\tau = 4,539.07$ €; it was then kept frozen), so that a ratio of two values — four evaluations, each in error by at most τ — is resolved to δ_R . The rule bounds the stopping error of each evaluation at about τ , not well inside it: on the case study ten of the certificates in use stop within 5 % of τ , and evaluations continued past certification moved by at most 0.9τ (Section 4.7).

Determinacy. A difference between two certified evaluations is called determinate when it exceeds $\max\{3b, 2\tau\}$, where b is the larger of the two certification bands (the range of the objective over the certifying window). A difference involving an uncertified evaluation is determinate only if it exceeds three times the larger of that evaluation’s consensus gap and settling slack. Every planning statement in this paper is a pairwise comparison of evaluations with this threshold stated, and the master search of Algorithm 2 accepts a poll point as an improvement only by a determinate margin; smaller differences are ties, reported as “within resolution”. What is certified is each evaluation, not the plan: the method claims no optimality certificate for the nonconvex recourse, and the bands, gaps and slacks of every evaluation used in a reported comparison are tabulated in B.

The storage schedule itself — charging, discharging and reactive power in every hour — is a consensus variable between the operators’ network models

and the shared-ESS agent: the network models carry the state-of-charge dynamics and the converter limits, and the agent carries the ageing chain that links the representative years. The two views are reconciled by the ADMM consensus channel on the storage schedule (A).

1.2.2 Installed and Available Capacity

The investment plan is a fixed parameter of the recourse. Each investment made at node e in year y^{Inv} forms a cohort that contributes its rated power and energy for its calendar lifetime ($\forall e \in E^S, \forall y^{\text{Inv}} \in Y, \forall y \in [y^{\text{Inv}}; y^{\text{Inv}} + T_{e,y^{\text{Inv}}}^{\text{Cal}}]$):

$$S_{e,y^{\text{Inv}},y}^{\text{Rated,Unit}} = \hat{S}_{e,y^{\text{Inv}}}^{\text{Inv}}, \quad E_{e,y^{\text{Inv}},y}^{\text{Rated,Unit}} = \hat{E}_{e,y^{\text{Inv}}}^{\text{Inv}} \quad (5)$$

The available power of a cohort equals its rating, while its available energy is reduced by its state of health at the start of the representative year ($\forall e, y^{\text{Inv}}, y$ as above):

$$S_{e,y^{\text{Inv}},y}^{\text{Av,Unit}} = S_{e,y^{\text{Inv}},y}^{\text{Rated,Unit}}, \quad E_{e,y^{\text{Inv}},y}^{\text{Av,Unit}} = E_{e,y^{\text{Inv}},y}^{\text{Rated,Unit}} SoH_{e,y^{\text{Inv}},y-1} \quad (6)$$

and the capacities seen by the network models in year y are the sums over the cohorts alive in that year ($\forall e \in E^S, \forall y \in Y$):

$$S_{e,y}^{\text{Av}} = \sum_{y^{\text{Inv}}=\max(y-T_{e,y}^{\text{Cal}}, y_0)}^y S_{e,y^{\text{Inv}},y}^{\text{Av,Unit}}, \quad E_{e,y}^{\text{Av}} = \sum_{y^{\text{Inv}}=\max(y-T_{e,y}^{\text{Cal}}, y_0)}^y E_{e,y^{\text{Inv}},y}^{\text{Av,Unit}} \quad (7)$$

1.2.3 Storage Operation in the Network Models

In every network model in which the shared ESS at node e appears, its operation in representative year y , day d and hour t is described by a charging power $P_{e,y,d,t}^{\text{Ch}} \geq 0$, a discharging power $P_{e,y,d,t}^{\text{Dch}} \geq 0$, a reactive power $Q_{e,y,d,t}^{\text{E}}$ and a stored energy $E_{e,y,d,t}^{\text{SoC}}$. The net active-power injection at the node is $P_{e,y,d,t}^{\text{E}} = P_{e,y,d,t}^{\text{Dch}} - P_{e,y,d,t}^{\text{Ch}}$ (positive when discharging), and the stored energy follows ($\forall e \in E^S, \forall y \in Y, \forall d \in D, \forall t \in T$):

$$E_{e,y,d,t}^{\text{SoC}} = E_{e,y,d,t-1}^{\text{SoC}} + \Delta t \left(\eta_e^{\text{Ch}} P_{e,y,d,t}^{\text{Ch}} - \frac{P_{e,y,d,t}^{\text{Dch}}}{\eta_e^{\text{Dch}}} \right) \quad (8)$$

$$SoC_e^{\text{Min}} E_{e,y}^{\text{Av}} \leq E_{e,y,d,t}^{\text{SoC}} \leq SoC_e^{\text{Max}} E_{e,y}^{\text{Av}} \quad (9)$$

$$E_{e,y,d,t_0}^{\text{SoC}} = E_{e,y,d,|T|}^{\text{SoC}} = SoC_e^0 E_{e,y}^{\text{Av}} \quad (10)$$

$$P_{e,y,d,t}^{\text{Ch}} \leq S_{e,y}^{\text{Av}}, \quad P_{e,y,d,t}^{\text{Dch}} \leq S_{e,y}^{\text{Av}}, \quad (P_{e,y,d,t}^{\text{E}})^2 + (Q_{e,y,d,t}^{\text{E}})^2 \leq (S_{e,y}^{\text{Av}})^2 \quad (11)$$

$$P_{e,y,d,t}^{\text{Ch}} P_{e,y,d,t}^{\text{Dch}} = 0 \quad (12)$$

where η_e^{Ch} and η_e^{Dch} are the charging and discharging efficiencies, SoC_e^{Min} , SoC_e^{Max} and SoC_e^0 the minimum, maximum and initial states of charge as fractions of the available energy, and Δt the hour length. Equation (10) closes every representative day on its initial state, so no energy is carried between representative days and cumulative discharge beyond the stored energy is excluded in every day and scenario by (8)–(10). Reactive power is a converter quantity: it is limited with the active power by the capability circle in (11) and has no effect on the stored energy or on ageing. The complementarity condition (12) is kept in the network models in its relaxed, penalized form as in [?]. In the multi-scenario instance the storage schedule of a block is a single scenario-independent variable referenced by every operation scenario's balance equations — a day-ahead commitment — so that the schedule the agent ages is the one every scenario runs (Subsection 2.3.4).

1.2.4 Degradation Chain of the Shared-ESS Agent

The shared-ESS agent holds the consensus copy of the storage schedule and ages the cohorts with it. Its schedule is allocated to the cohorts alive at node

e in proportion to their rated energy — a homogeneous-fleet approximation, in which all cohorts at a node cycle at the same depth ($\forall e, y^{\text{Inv}}, y, d, t$):

$$P_{e,y^{\text{Inv}},y,d,t}^{\text{Ch}} = \frac{E_{e,y^{\text{Inv}},y}^{\text{Rated,Unit}}}{\sum_{y'} E_{e,y',y}^{\text{Rated,Unit}}} P_{e,y,d,t}^{\text{Ch}}, \quad P_{e,y^{\text{Inv}},y,d,t}^{\text{Dch}} = \frac{E_{e,y^{\text{Inv}},y}^{\text{Rated,Unit}}}{\sum_{y'} E_{e,y',y}^{\text{Rated,Unit}}} P_{e,y,d,t}^{\text{Dch}} \quad (13)$$

Ageing is driven by the energy that crosses the cells, not by the apparent power at the converter: the average daily cell-side throughput of a cohort in representative year y is

$$\bar{E}_{e,y^{\text{Inv}},y}^{\text{Cell}} = \sum_{d \in D} \frac{D_d}{365} \sum_{t \in T} \Delta t \left(\eta_e^{\text{Ch}} P_{e,y^{\text{Inv}},y,d,t}^{\text{Ch}} + \frac{P_{e,y^{\text{Inv}},y,d,t}^{\text{Dch}}}{\eta_e^{\text{Dch}}} \right) \quad (14)$$

and the fractional cycling loss accumulated over the Y_y calendar years of the block is

$$D_{e,y^{\text{Inv}},y} = \frac{365 Y_y \bar{E}_{e,y^{\text{Inv}},y}^{\text{Cell}}}{2 k_e E_{e,y^{\text{Inv}},y}^{\text{Rated,Unit}}} \quad (15)$$

where k_e is the effective cycle life of the technology: $2E^{\text{Rated,Unit}}$ is the throughput of one full cycle, so D counts equivalent full cycles against k_e . The state of health then follows a continuous-compounding chain with a calendar retention factor ϕ_e^{Cal} per calendar year ($\forall e, y^{\text{Inv}}, y$):

$$SoH_{e,y^{\text{Inv}},y} = SoH_{e,y^{\text{Inv}},y-1} \exp(-D_{e,y^{\text{Inv}},y}) (\phi_e^{\text{Cal}})^{Y_y}, \quad SoH_{e,y^{\text{Inv}},y^{\text{Inv}}-1} = 1 \quad (16)$$

$$SoH_{e,y^{\text{Inv}},y} \geq SoH_e^{\text{Min}} \quad (17)$$

The constant k_e is calibrated from a datasheet statement of N^{DS} cycles at depth of discharge δ^{DS} to a retention R^{DS} of the rated capacity: with (15)–(16) and no calendar term, those cycles give $SoH = \exp(-\delta^{\text{DS}} N^{\text{DS}}/k_e) = R^{\text{DS}}$, hence

$$k_e = \frac{\delta^{\text{DS}} N^{\text{DS}}}{-\ln R^{\text{DS}}} \quad (18)$$

The calibrations used, the calendar retention and the end-of-life floor SoH_e^{Min} are given in Section 3.4. The floor is the only intertemporal trade-off enforced in operation: no explicit cycling cost is charged, because a shared asset that has been paid for carries no operating cost of its own; degradation is tracked and propagated, and its economic effect appears through the available energy of later years and through the floor. Equations (15)–(17) are linear in D and SoH for a fixed schedule, and (16) is the only nonlinear row of the agent.

1.2.5 Objective of the Shared-ESS Agent

The agent has no economic objective. Within the coordination procedure it minimizes the augmented-Lagrangian terms of the storage consensus channel (A.2) plus a small linear regularization of throughput,

$$\begin{aligned} \min \varepsilon^E \sum_{e \in E^S} \sum_{y^{\text{Inv}} \in Y} \sum_{y \in Y} \sum_{d \in D} \sum_{t \in T} (P_{e,y^{\text{Inv}},y,d,t}^{\text{Ch}} + P_{e,y^{\text{Inv}},y,d,t}^{\text{Dch}}) \\ + \sum_{e \in E^S} \sum_{y \in Y} \sum_{d \in D} \sum_{t \in T} (\mathcal{L}_{e,y,d,t}^{E,P} + \mathcal{L}_{e,y,d,t}^{E,Q}) \end{aligned} \quad (19)$$

subject to (5)–(7), (13)–(17), $0 \leq P_{e,y^{\text{Inv}},y,d,t}^{\text{Ch}}, P_{e,y^{\text{Inv}},y,d,t}^{\text{Dch}} \leq S_{e,y^{\text{Inv}},y}^{\text{Rated,Unit}}$ and the converter capability inequality of (11) on the aggregate schedule. The regularization weight ε^E is small enough to be below the resolution of Subsection 2.2.7 and serves one purpose: with a linear, strictly positive price on $P^{\text{Ch}} + P^{\text{Dch}}$, simultaneous charging and discharging is never optimal, so complementarity in the agent follows from the problem structure and needs no nonconvex constraint. The maximum residual $\min\{P^{\text{Ch}}, P^{\text{Dch}}\}$ is checked after every solve and reported. The earlier investment-fixing slacks and the apparent-power aggregation are retired: investments are parameters, and reactive power enters only through the capability circle.

1.2.6 Day-Ahead Commitment under Operation Scenarios

In the multi-scenario instance each operation scenario o is instantiated inside every network block with its own load and RES realization, while the interface schedule committed to the TSO is scenario-independent by construction

in the TSO model and may be deviated from in the DSO model. Each DSO carries a day-ahead expectation of its interface flow and is charged for deviating from it. With $P_{i,o,t}^I$ the interface active power of DSO i in scenario o and hour t (likewise $Q_{i,o,t}^I$), the expectation and the deviations are $(\forall i, o, t)$:

$$\bar{P}_{i,t}^I = \sum_{o \in \Omega_O} \omega_o P_{i,o,t}^I, \quad P_{i,o,t}^I - \bar{P}_{i,t}^I = d_{i,o,t}^+ - d_{i,o,t}^-, \quad d_{i,o,t}^\pm \geq 0 \quad (20)$$

and the charge added to the DSO's objective, and hence to C^{Op} in (4), is

$$\sum_{o \in \Omega_O} \sum_{t \in T} \omega_o \alpha \bar{\pi}_t (d_{i,o,t}^+ + d_{i,o,t}^-) \quad (21)$$

where $\bar{\pi}_t$ is the probability-weighted wholesale price of the hour over the market scenarios and α an imbalance premium, so that a deviation is priced at a fraction α of the energy price of the hour in which it occurs. The same term is written on the reactive interface flow with the same price, as a stated assumption. The TSO operates on the committed schedule and carries no deviation term; the interface voltage carries none either, being a physical state rather than a commitment. The storage schedule is not deviated from: it is one scenario-free variable per hour in every block (Subsection 2.3.2), so the committed schedule is the one that ages the cells. At a single operation scenario (20)–(21) vanish identically, and the single-scenario instance is unaffected by them. The value of α is given in Section 3.5.

A A

A.1 A1

A.2 A2

B E